

WeekB Homework

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P311 T2

Prove that the following polynomials are irreducible in $\mathbb{Z}[x]$:

b. $x^6 + 30x^5 - 15x^3 + 6x - 120$.

d. $\frac{(x+2)^p - 2^p}{x}$, where p is an odd prime.

b. We have $f(x) = x^6 + 30x^5 - 15x^3 + 6x - 120$, then $f'(x) = f(x) \bmod 9 = x^6 + 3x^5 + 3x^3 - 3x - 3$. Since $\mathbb{Z}_9/(3) = \mathbb{Z}_3$ is a field, so (3) is prime in $\mathbb{Z}_9$. Since $f'_6 = 1$, $f'_i \in (3)$, $f'_0 \notin (3)^2 = (0)$, $f'(x)$ thus $f(x)$ is irreducible.

d. We have $g(x) = \frac{(x+2)^p - 2^p}{x} = \sum_{i=0}^{p-1} \binom{p}{i+1} x^i$, so $g_{p-1} = 1$, $p \mid g_i$, $p^2 \nmid g_0$, thus $g(x)$ is irreducible.

P311 T7

Prove that $R[x]/(x^2 + 1)$ is a field which is isomorphic to the complex numbers.

Easy to verify $(x - a)(x - b) \neq x^2 + 1$ when $a, b \in R$. So $x^2 + 1$ is irreducible and belongs to $R[x]$. So $K = R[x]/(x^2 + 1)$ is a extend of R , $\{1, \theta\}$ is a basis for K . So $K = \{a + b\theta \mid a, b \in R\}$, and we have $\theta^2 + 1 = 0$, that is $\theta^2 = -1$. So $K = \mathbb{C}$.

P312 T18

Show that $6x^5 + 14x^3 - 21x + 35$ and $18x^5 - 30x^2 + 120x + 360$ are irreducible in $\mathbb{Q}[x]$.

$6x^5 + 14x^3 - 21x + 35$: We have $7 \nmid 6$, $49 \nmid 35$, but $7 \mid 14, 21, 35$. Then irreducible in $\mathbb{Z}[x]$, thus in $\mathbb{Q}[x]$ by Gauss.

$18x^5 - 30x^2 + 120x + 360$: Likewise, use $p = 5$.

P312 T20

Show that the polynomial $f(x) = x$ in $\mathbb{Z}/6\mathbb{Z}[x]$ factors as $(3x + 4)(4x + 3)$, hence is not an irreducible polynomial.

a. Show that the reduction of $f(x)$ modulo both of the nontrivial ideals (2) and (3) of $\mathbb{Z}/6\mathbb{Z}$ is an irreducible polynomial, showing that the condition that R be an integral domain in Proposition 12 is necessary.

b. Show that in any factorization $f(x) = g(x)h(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$ the reduction of $g(x)$ modulo (2) is either 1 or x and the reduction of $h(x)$ modulo (2) is then either x or 1, and similarly for the reductions modulo (3). Determine all the factorizations of $f(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$.

c. Show that the ideal $(3, x)$ is a principal ideal in $\mathbb{Z}/6\mathbb{Z}[x]$.

d. Show that over the ring $\mathbb{Z}/30\mathbb{Z}[x]$ the polynomial $f(x) = x$ has the factorization $f(x) = (10x + 21)(15x + 16)(6x + 25)$. Prove that the product of any of these factors is again of the same degree.

Prove that the reduction of $f(x)$ modulo any prime in $\mathbb{Z}/30\mathbb{Z}$ is an irreducible polynomial. Determine all the factorizations of $f(x)$ in $\mathbb{Z}/30\mathbb{Z}[x]$.

e. Generalize part d. to $\mathbb{Z}/n\mathbb{Z}[x]$ where n is the product of k distinct primes.

a. We need to prove $f(x) = x$ is irreducible in $\mathbb{Z}/2\mathbb{Z}[x]$ and $\mathbb{Z}/3\mathbb{Z}[x]$. Since $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ is field, we prove $f(x) = x$ is irreducible in $F[x]$. If $f = ab$, we have $a_0b_0 = 0$, thus suppose $a_0 = 0$. Thus f irreducible iff $g(x) = 1$ irreducible. And we know 1 is prime in $F[x]$, and $F[x]$ is ED.

b. Suppose $a_0 = 0$, we have $a_1b_0 = 1$. Thus $a_1 = b_0 = 1$. Take max n, m that $a_n = b_m = 1$, so $f_{n+m} = 1$. Thus $n + m = 1$, so only $a(x) = x, b(x) = 1$. Likewise, $a(x) = 1, b(x) = x$.

If mod 3, then the coefficient of a, b can both be 2.

If mod 6, that is $x * 1, (3x + 4)(4x + 3), 5x * 5, (3x + 2)(2x + 3)$, and diverse by comm.

c. We have $(3, x) = (4x + 3)$. Since $4x + 3 \in (3, x), (3x + 4)(4x + 3) = x, 3(4x + 3) = 3$.

d. Mod 2: 1, x , 1. Mod 3: x , 1, 1. Mod 5: 1, 1, x . So the mul is x mod 30.

Any two: Equal to x mod 2 primes, and equal to 1 mod the other. So the mul of two term is a 1-deg poly.

We already prove $f(x)$ mod any prime is irreducible in a.

All factorization is showed in e. as a special case.

e. Let $n = p_1 \cdots p_k$. Pick $n_1, n_2, \dots, n_k$ that $n_i \equiv [i = j] \pmod{p_j}$ by CRT. For all $i \in [1, k]$, pick $a_{i1}, a_{i2}, \dots, a_{ik}$ that $\prod_j a_{ij} \equiv 1 \pmod{p_i}$. All factorization is in form of $f = g_1g_2 \cdots g_k$, where $g_j = n_j a_{jj}x + \sum_{i \neq j} n_i a_{ij}$. Since $g_j \equiv 1 \pmod{p_i}$ for $i \neq j$, $g_j \pmod{p_j}$ is irreducible, then g_j is irreducible.

P315 T3

Let p be an odd prime in $\mathbb{Z}$ and let n be a positive integer. Prove that $x^n - p$ is irreducible over $\mathbb{Z}[i]$.

Since $\mathbb{Z}[i]$ is ED, we have p is prime when $p \equiv 3 \pmod{4}$ and otherwise $p = a^2 + b^2$ unique, $a + bi$ and $a - bi$ is prime.

If p is prime, then $x^n - p$ is irreducible since $p \notin (p)^2$.

Otherwise, we have $N((a + bi)^2) = N(a + bi)^2 = N(p)$. Since $p \neq (a + bi)^2$, and all 1-norm element is unit, we have $p \notin (a + bi)^2$, thus $x^n - p$ is irreducible.

P315 T7

Prove that the additive and multiplicative groups of a field are never isomorphic.

If $-1 \neq 1$ in F , We have $|-1| = 2$ in $F^\times$. If $x + x = 0$ in F^+ , then $x = -x$, $1 = -1$.

If $-1 = 1$ in F , We have $|x| = 2$ for $x \neq 0$ in F^+ . So we have $x^2 = 1$ for all $x \in F^\times$. But $x^2 = 1$ has atmost 2 solution, thus $F^\times$ is finite. $|F^+| = |F|$ but $|F^\times| = |F| - 1$.

P519 T1

Show that $p(x) = x^3 + 9x + 6$ is irreducible in $\mathbb{Q}[x]$. Let θ be a root of $p(x)$. Find the inverse of $1 + \theta$ in $\mathbb{Q}(\theta)$.

Since $6 \notin (3)^2$, $9, 6 \in (3)$, $p(x)$ is irreducible in $\mathbb{Z}[x]$, thus is irreducible in $\mathbb{Q}[x]$ by Gauss.

We have $\theta^3 + 9\theta + 6 = 0$, so easy to verify $(1 + \theta)^{-1} = \frac{1}{4}(10 - \theta + \theta^2)$.

P519 T2

Show that $x^3 - 2x - 2$ is irreducible over $\mathbb{Q}$ and let θ be a root. Compute $(1 + \theta)(1 + \theta + \theta^2)$ and $\frac{1+\theta}{1+\theta+\theta^2}$ in $\mathbb{Q}[\theta]$.

Since $2 \notin (2)^2$, $2 \in (2)$, the poly is irreducible.

We have $\theta^3 - 2\theta - 2 = 0$. So $(1 + \theta)(1 + \theta + \theta^2) = 3 + 4\theta + 2\theta^2$. $\frac{1+\theta}{1+\theta+\theta^2} = \frac{1}{3}(1 + 2\theta - \theta^2)$.

P519 T5

Suppose a is a rational root of a monic polynomial in $\mathbb{Z}[x]$. Prove that a is an integer.

Let $a = \frac{r}{s}$, $p(x)$ is the poly mentioned, we have $s \mid p_n = 1$, so $s = 1$.

P519 T6

Show that if α is a root of $a_n x^n + \dots + a_0$ then $a_n \alpha$ is a root of the monic polynomial $x^n + a_{n-1} x^{n-1} + a_n a_{n-2} x^{n-2} + \dots + a_n^{n-1} a_0$.

We have $a_n^n \alpha^n + a_n^{n-1} a_{n-1} \alpha^{n-1} + \dots + a_n^{n-1} a_0 = 0$. So $(a_n \alpha)^n + a_{n-1} (a_n \alpha)^{n-1} + \dots + a_n^{n-1} a_0 = 0$.